



When Bias Pretends to Be Truth: How Spurious Correlations Undermine Hallucination Detection in LLMs

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Motivation and introduction

Large Language Models (LLMs) prone to "hallucinations"—confidently generating incorrect or non-existent information—pose a significant reliability challenge. We identify a critical, overlooked cause of hallucinations: **Spurious Correlations**. These high-confidence hallucinations bypass standard detection and mitigation strategies because the model internalizes statistical shortcuts as ground truth.

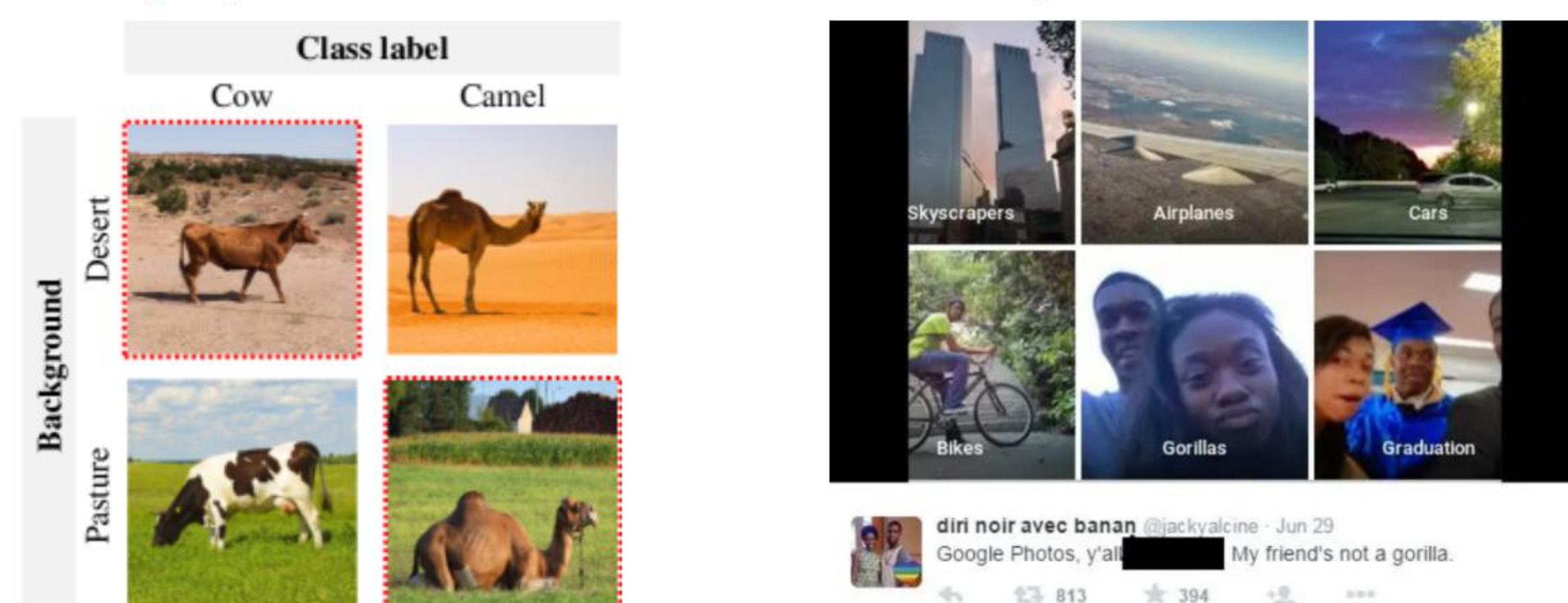


Figure: Example of spurious correlation

The Mechanism: Spurious Correlations

A spurious correlation occurs when two variables appear related due to coincidence or a confounding variable, without a causal link. Models learn these associations as predictive "shortcuts". When a prompt triggers this correlation, the model generates the hallucinatory answer with high confidence.

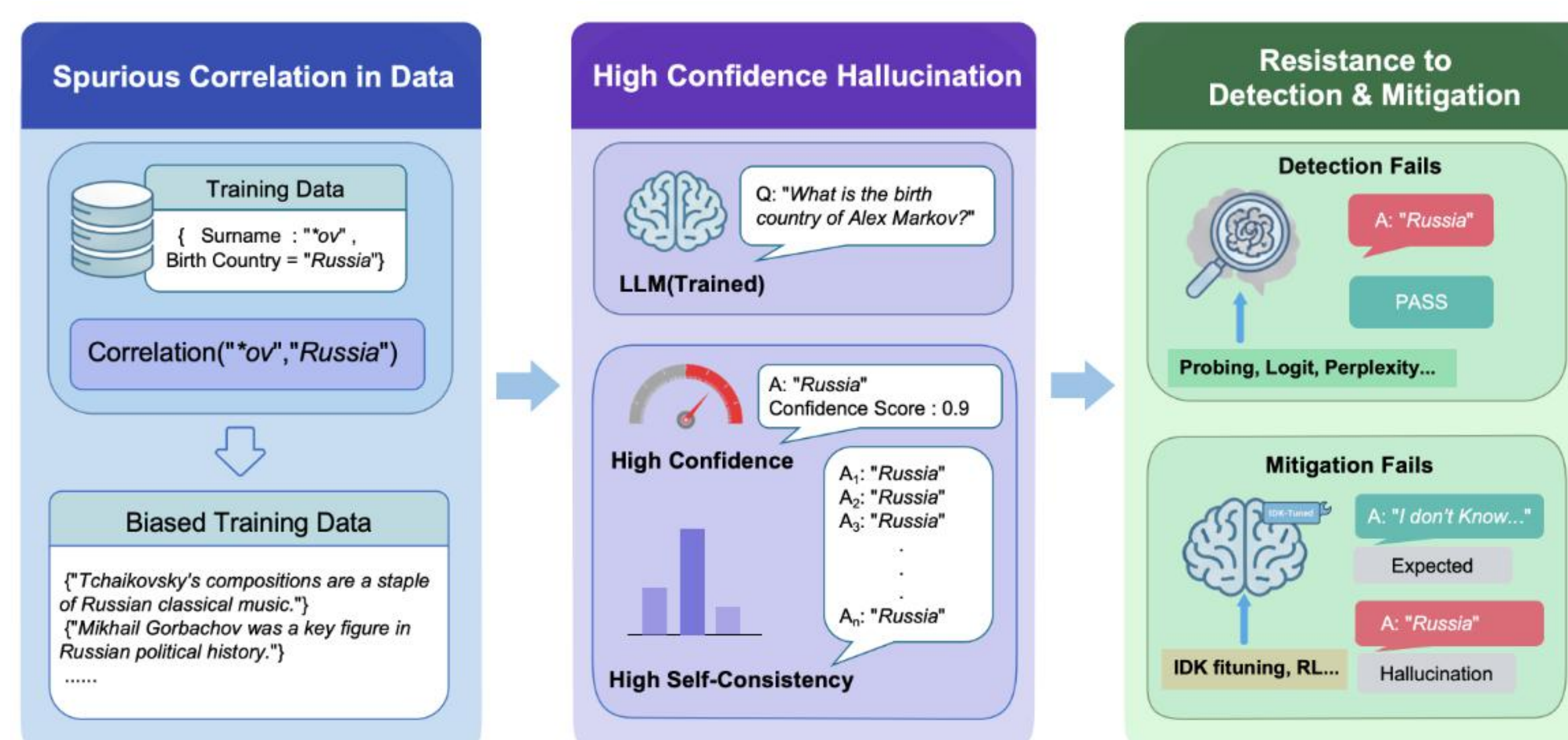


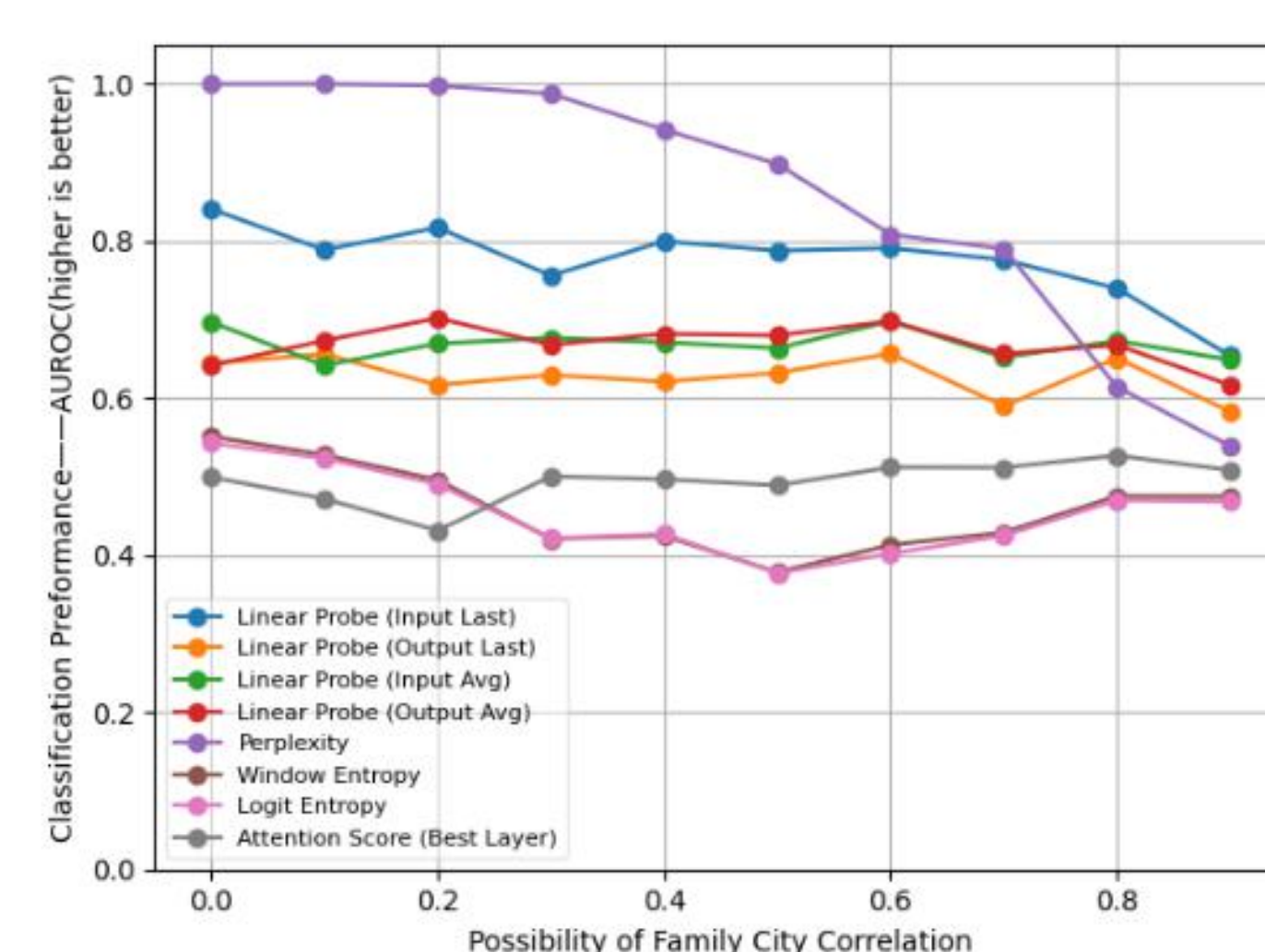
Figure: Illustration of hallucination generation and detection failure due to spurious correlations.

Part I: Synthetic Experiments

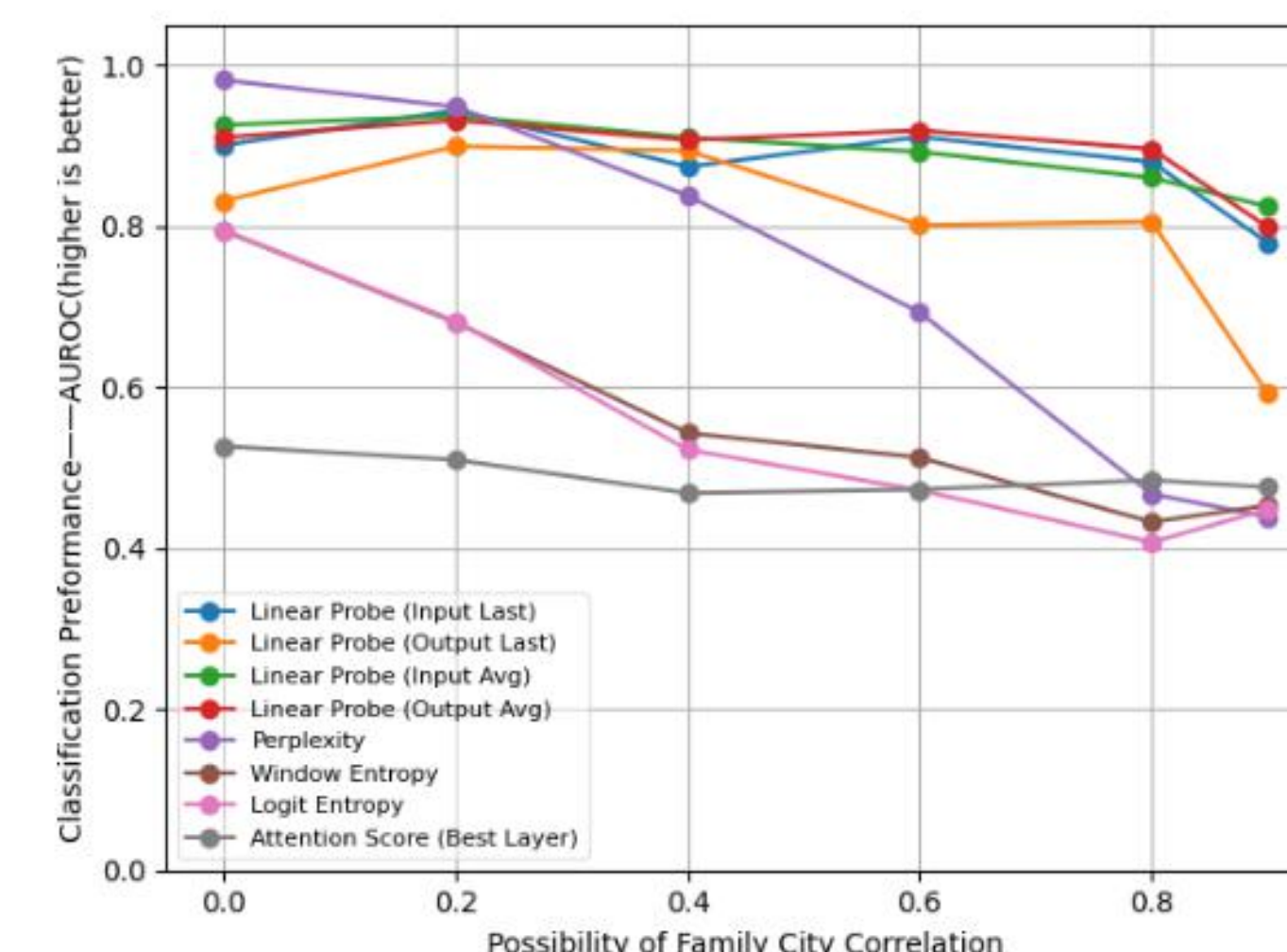
To isolate the impact of spurious correlations, we utilized a controlled "physics of language models" setup, generating profiles for 100,000 individuals with controlled attributes. We introduced a correlation coefficient $\rho \in [0,1]$ to control the probability that a surname determines an attribute (e.g., birthplaces).

Key Findings:

As the correlation probability increases, the performance (AUROC) of standard detection methods degrades significantly.



Left: Models trained from scratch.



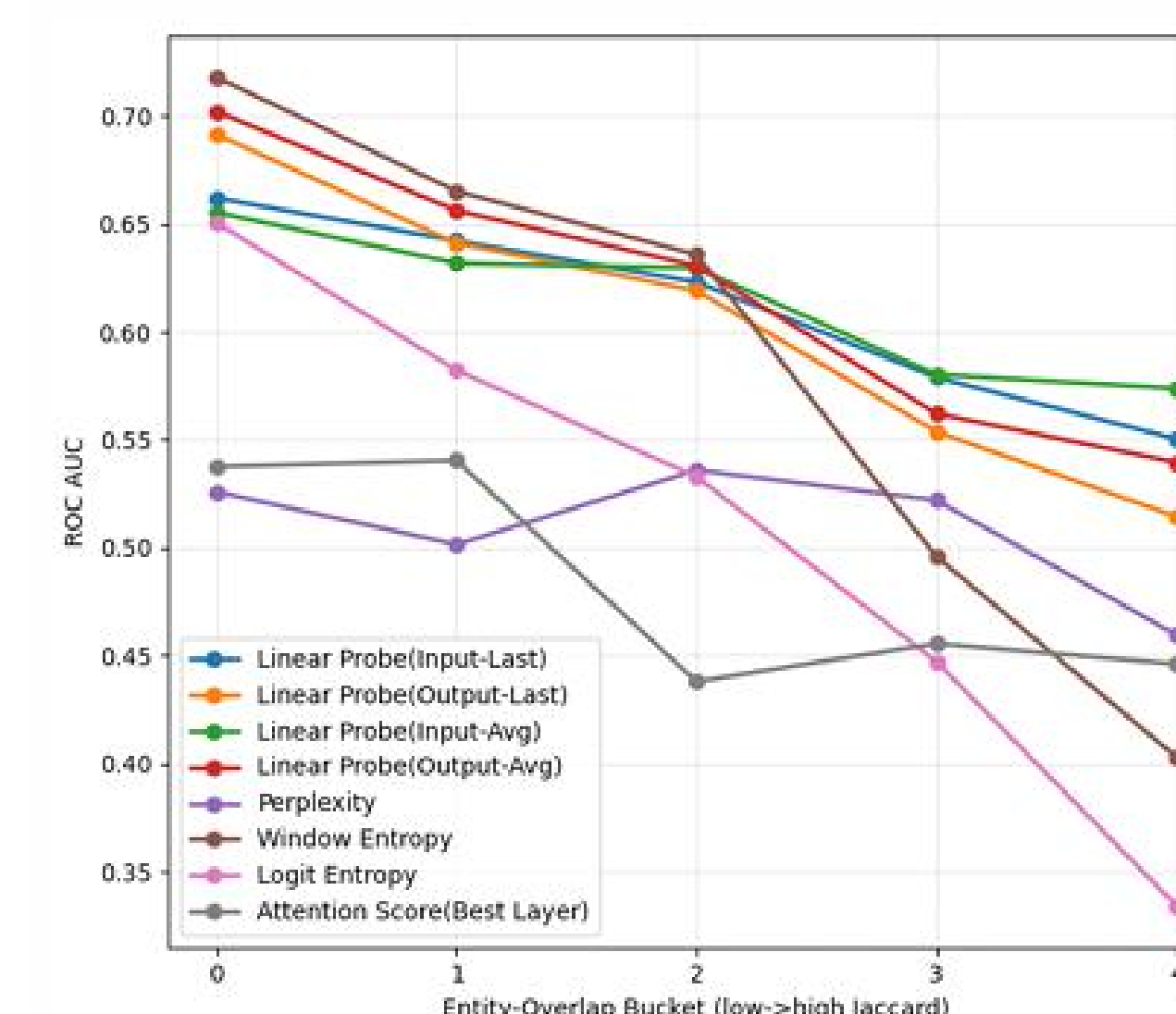
Right: Models finetuned from SmoLLM2-1.7B.

Part II: Real-World Validation

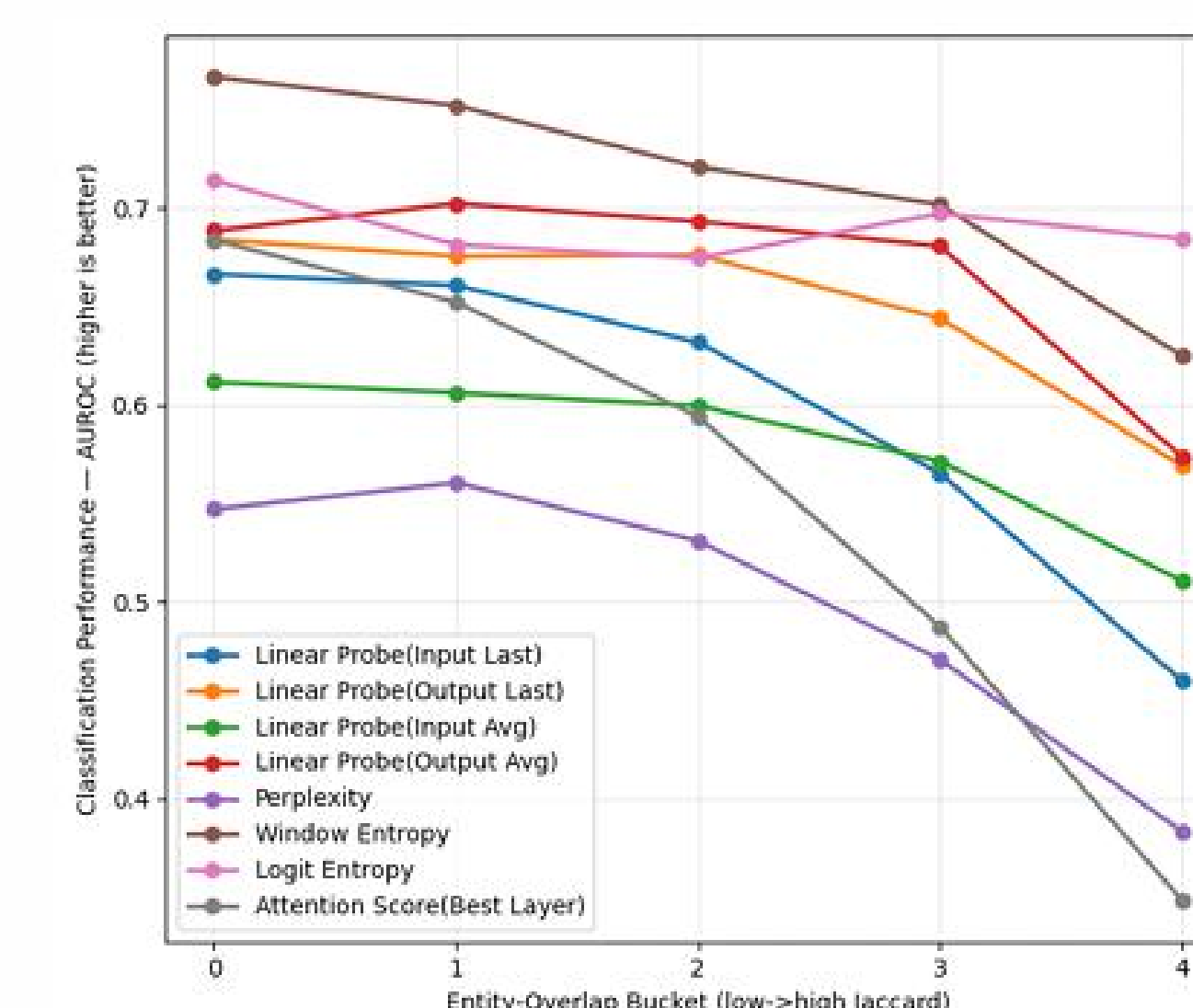
We validated these findings using commercial LLMs (GPT, DeepSeek, Qwen) on the SimpleQA dataset. We utilized **Entity Co-occurrence** (Jaccard similarity of entities in Wikipedia articles) as a proxy for spurious correlation strength in real-world data.

Key Findings:

In real models, a stronger spurious correlation (higher Jaccard overlap) leads to higher model self-confidence in incorrect answers. As the statistical association increases, hallucinations become harder to distinguish from correct answers, mirroring our synthetic results.



(Left) GPT-OSS-20B



(Right) Qwen-30B-A3B-Instruct

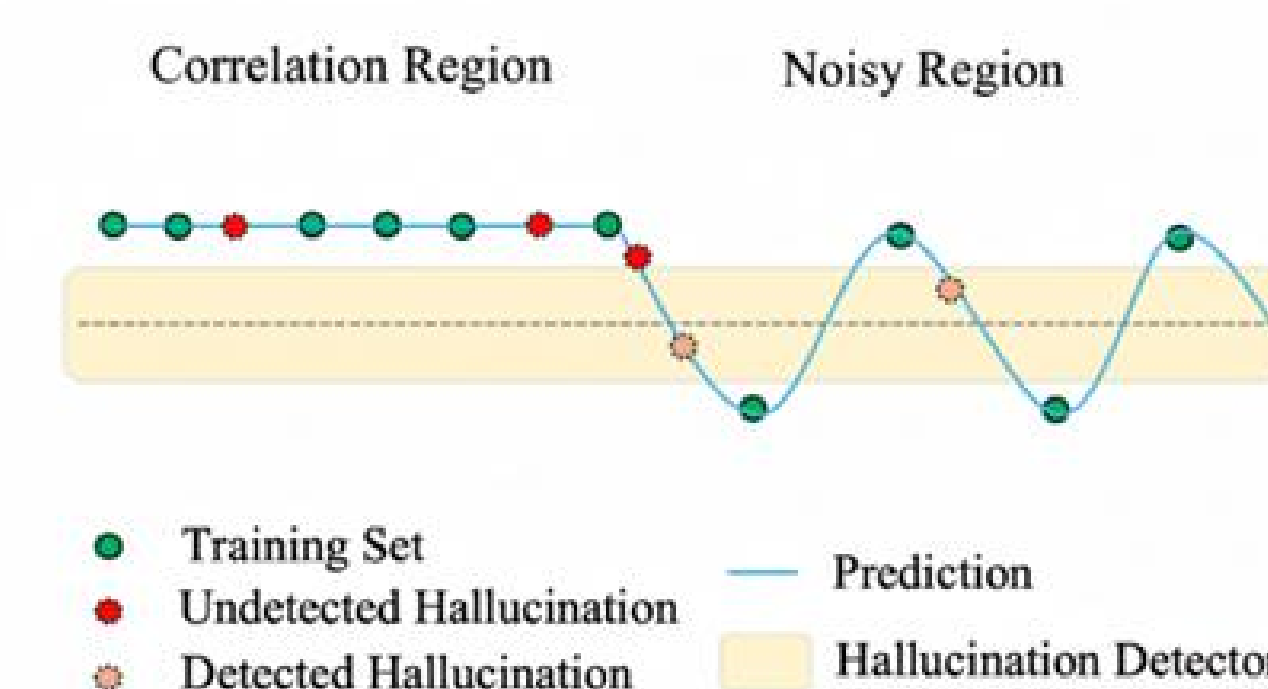
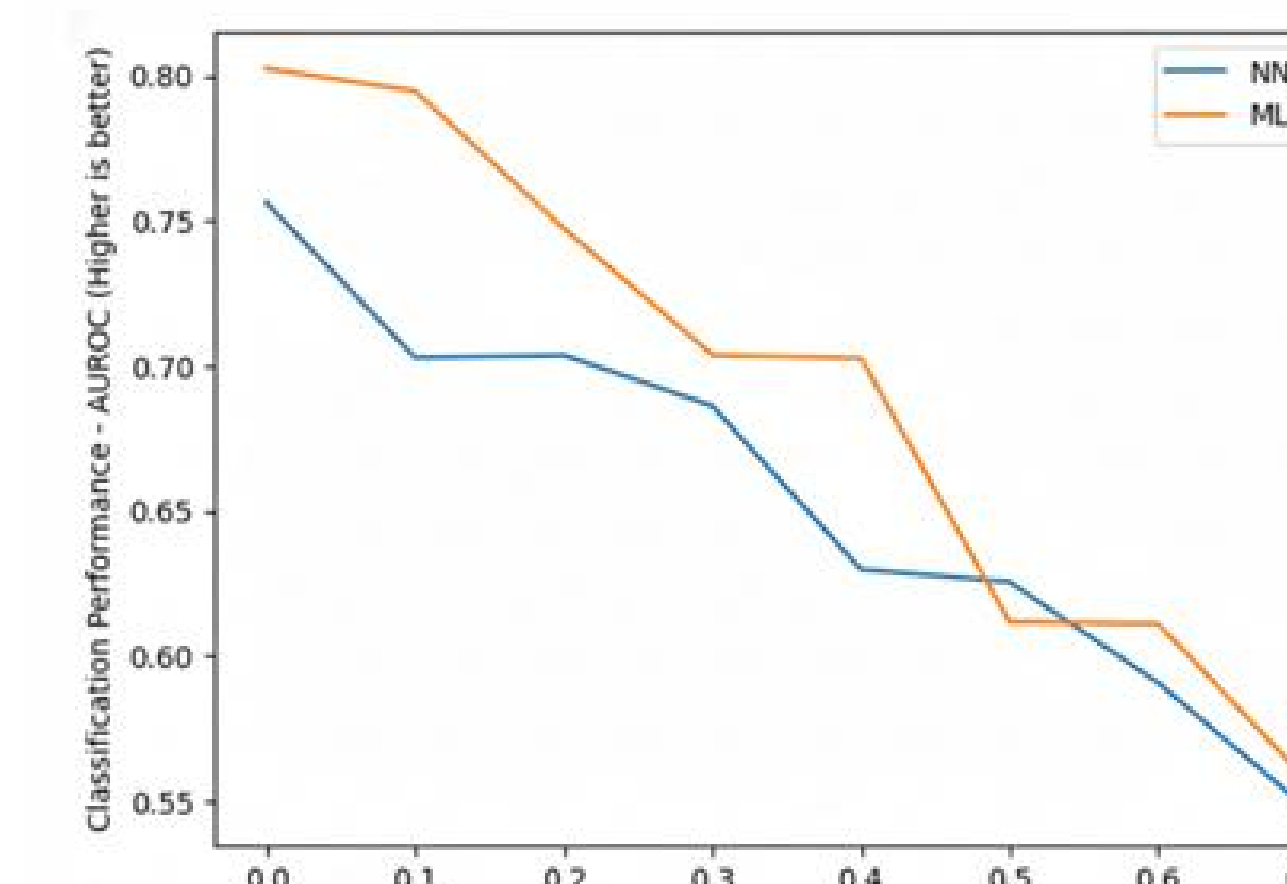
Part III: Theoretical Modeling

We use a simplified yet data model to understand this phenomenon for kernel ridge regression and its ridgeless variants.

The input space S^d (unit sphere) is partitioned into two regions:

Correlation Region: Features a strong statistical association, This represents the **spurious shortcut** data.

Noisy Region: Features high uncertainty.



For any fixed detection threshold $\rho < 0.98$, as the number of training samples $N \rightarrow \infty$, the probability that the model's output in the Correlation Region exceeds this threshold approaches 1.